closed capability inside a component supplied through an apparently compliant vendor. Indeed, industry sources have raised concerns about Chinese-origin components being routed through third country brands (that is, 'white-labelled') to meet origin requirements.

The greatest weakness actually lies several tiers below the prime contractor. A defence company may know its immediate suppliers, but those suppliers may themselves depend on subcontractors for chips, sensors, radio modules, firmware and other components. By the time the supply chain reaches the third or fourth tier, visibility can become extremely limited.

STRUCTURAL WEAKNESS

Major General Pawan Anand (retd), director, Centre for Emerging Technology at the United Service Institution of India, identifies this as a critical gap in India's current approach. "Tender documents require vendors to disclose component origin at the point of bidding, but an audit trail that follows a component through sub-suppliers once the contract is signed is difficult to establish," he says. A drone can be assembled in Bengaluru and still contain a chipset, a radio module or firmware written by a fourth-tier vendor whom nobody in Delhi has heard of. "That is the structural weakness in 'indigenous' labelling," Anand says.

An Indian company may label a platform 'indigenous', yet along with its prime supplier of parts, be in the dark about the exact capabilities of a foreign component. As Indian firms move towards manufacturing all components, they must in the meantime urgently develop the ability to independently understand and control security-critical technology. "There is a crucial distinction between manufacturing sovereignty and verification sovereignty," Brar says. "If you cannot make it, you must at least be able to verify it, control it and replace it."

Beyond country-of-origin vetting, procurement agencies of military drones need to establish what their critical components can do, what external connections they can establish, how they receive updates and whether those functions can be controlled by the military user. "'Indigenous' has become a procurement category more than a security guarantee and the gap between the two is where incidents like the K3 Scout come from," Maj Gen. Anand says. Calling a drone secure is just "describing an intention and not a fact" until India can trace a component's behaviour, he adds.

THE VULNERABLE PARTS

Security-critical components in a military drone

- **Flight-control software/ firmware/ autopilot:** As the digital brain of the drone, it is the critical cyber-physical layer responsible for its functioning
- **Communications and datalink:** Vulnerability here can affect command, control, telemetry and mission data
- **GNSS (Global Navigation Satellite System) and navigation software:** Spoofing/ jamming can alter a drone's navigation. Most have safeguards like a 'return to home' feature
- **Mission Computer / System-on-Chip (SoC):** Integration between hardware (SoC) and mission planning software (mission computer) can be a source of weakness
- **Infrared cameras/ sensors:** Modern sensors with processing, networking and firmware capabilities can be hit too
- **Ground-control station:** The central command hub used by a drone operator on the ground needs to be watched closely

UAVS THE

Some of the indigenous drones with the Indian armed forces



HAWK & SCOUT
Surveillance/ early warning drones from Zuppa Geo Navigation Technologies

Role: Provided real-time intel into enemy territory during Operation Sindoor

ALS-50 & JM-1
Kamikaze (weaponised fire-and-forget) drones from Tata Advanced Systems Ltd and Johnnette Technologies respectively. Carries anti-personnel and anti-armour warhead

Role: Made combat debut in Op. Sindoor

NAGASTRA-1
Kamikaze drones made by Solar Industries; carries high



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